

Dao Xuan Tan

Data Scientist

📍 Ho Chi Minh, Vietnam

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🌐 Đào Xuân Tân

Summary

Data Scientist building and evaluating ML and Deep Learning models in production, from training and fine-tuning to monitoring for degradation. Experienced with LLM and AI-agent systems, grounded in a strong probability, statistics, and linear algebra foundation from a Mathematics and Computer Science (Data Science) degree, plus production Python and SQL.

Experience

AcamarAI, AI Engineer

ML/DL modeling, LLM agents, model evaluation, PyTorch, vLLM, RAG, FastAPI

Ho Chi Minh

June 2025 – Jan 2026

8 months

- Built, fine-tuned, and shipped task-specific **ML and Deep Learning** models with **PyTorch** and **LoRA**, preparing datasets and optimizing inference quality for production use cases
- Designed **model evaluation and testing workflows**, measuring quality with **recall@k** and **LLM-as-judge** harnesses plus human-in-the-loop review, gating models before and after deployment
- Developed **agentic LLM** systems on **OpenAI tool calling** and **LangChain** with multi-step planning, enforcing schema-validated outputs via **Instructor/Pydantic** for reliable tool use
- Built production **RAG and retrieval** over **millions of vectors** in self-hosted **Qdrant**, combining **hybrid** dense and sparse (BM25) search with a **reranking** model that beat pure vector search on recall@k, plus **ColPali** vision RAG over document images
- Served self-hosted **Qwen** models with **vLLM** on a single A100 80GB behind an **OpenAI-compatible API**, and shipped **3 of 14** core AI features for Fortune 500 beta users via **async FastAPI** with **SSE** streaming

Pi Associates, Software Developer (Quant Finance)

Python, SQL, Pandas, PostgreSQL, Kafka, Redis, Docker, Kubernetes, GCP, Grafana

Ho Chi Minh

Jan 2026 – present

7 months

- Main developer of an internal **Python** data and backtesting platform used by about 20 researchers, running **tens of thousands** of backtests a day on **1 to 5 GB** of data per run
- Diagnosed and fixed a silent production **memory leak**, cutting peak memory from **200+ GB to under 20 GB**
- Reduced backtest runtime from **minutes to under 10 seconds** through **parallelism**, vectorization, and algorithmic improvements
- Worked across **PostgreSQL** (schema design and **SQL** queries), **Redis** caching, and a **Kafka** producer feeding downstream consumers
- Ran jobs as Dockerized **Kubernetes** workloads across **GCP** and on-premise, with **GitHub Actions** CI/CD and **Grafana** monitoring to track performance and catch degradation, plus stronger error handling and validation to reduce failed runs

Skills

Machine Learning & Deep Learning: PyTorch, deep learning, LoRA fine-tuning, model training and evaluation, recall@k, LLM-as-judge, production model monitoring, dataset preparation

Math & Statistics: probability, statistics, linear algebra, applied mathematics

LLM, GenAI & Agents: RAG, retrieval, tool/function-calling agents, multi-step planning, LangChain, prompt engineering, vLLM, self-hosted LLM serving, OpenAI-compatible APIs

Programming & Data: Python, SQL, PostgreSQL, FastAPI, Pandas, Kafka, Redis, data pipelines, REST APIs, Pydantic validation

MLOps & Infra: Docker, Kubernetes, GCP, on-premise, GitHub Actions CI/CD, Grafana

Languages: Vietnamese (Native), English (IELTS 6.5)

Education

Ho Chi Minh University of Science, Mathematics & Computer Science (Data Science)

Ho Chi Minh City

- GPA: 8.74/10.0 (3.74/4.0)

Oct 2022 – Dec 2026